ALGEBRAIC THEORY OF MODULAR PROJECTION SIEVING: PRIME-COPRIME ENTANGLEMENT AND ASYMMETRIC ACTIVATION THRESHOLDS

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ABSTRACT. We develop an algebraic theory of modular projection sieving over the unit group $(\mathbb{Z}/6\mathbb{Z})^\times$. The two active channels are $\mathcal{C}_1 = \{6k+1\}$ and $\mathcal{C}_5 = \{6k-1\}$. For every base prime $p = 6k_p+\varepsilon_p > 3$ we formalise a *positional coprime* $\mathrm{Cop}(p)$ in the opposite channel, and we prove that the correct activation threshold of p in $\mathcal{C}_5$ is $\kappa^-_{\min}(p) = p\,\mathrm{NextKop}(p) - \varepsilon_p\,k_p$, where $\mathrm{NextKop}(p)$ is the positional index of $\mathrm{Cop}(p)$. This asymmetric threshold strictly dominates the symmetric wheel threshold $\kappa^+_{\min}(p) = (p^2-1)/6$, with equality only in the twin-prime case. We call the resulting chiral pairing of base primes *prime-coprime entanglement*.

The entanglement structure identifies the vanishing of the entanglement jump $\Delta k(p) = \mathrm{NextKop}(p) - k_p$ with the local ground state of the prime-coprime pairing, and characterises the indices that generate twin primes via a quadratic annihilator over $(\mathbb{Z}/6\mathbb{Z})^\times$. Under the Hardy–Littlewood independence hypothesis, the local product $\prod_{p\geq 5}(1-(p-1)^{-2}) = \frac{4}{3}C_2$ arises as the survival density of the annihilator.

The resulting sieve algorithm runs in $\Theta(\sqrt{N}/\log N)$ working memory and $O(N^{3/2}/\log N)$ time, matching the lower bound for state-vectorless positional sieves. We report exact counts of $\pi(N)$ with wall-clock timings under a strict measurement protocol (warm-up, median over ten repetitions, bootstrap 95% confidence intervals) up to $N \sim 3\times 10^8$, and the entanglement jump distribution at two scales with Wilson 95% confidence intervals. Empirical transition ratios match the theoretical prediction to within 5% for $N \geq 10^6$. All algebraic results of §2–3 and §6.2 are certified in Lean 4 with zero `sorry` gaps and zero global axioms (nineteen modules: seventeen at Level N1 and two at Level N1-P).

1. INTRODUCTION

The wheel sieve of Pritchard [1, 2] exploits the unit group $(\mathbb{Z}/m\mathbb{Z})^\times$ to eliminate, before any arithmetic test, all integers whose residue modulo m

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All algebraic results of the manuscript are certified in Lean 4 [13] using Mathlib4 [14] with zero `sorry` gaps and zero global axioms (nineteen modules: seventeen at Level N1 and two at Level N1-P); see Appendix D..

is divisible by a prime divisor of m. For $m = 6$ the unit group is $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$ and the sieve operates on the two channels

$$\mathcal{C}_1 = \{6k+1\}_{k\geq 1}, \qquad \mathcal{C}_5 = \{6k-1\}_{k\geq 1}, \tag{1.1}$$

eliminating exactly 2/3 of integers and retaining all primes $p > 3$. The surviving candidates are then tested against a base prime table of size $\pi(\sqrt{N})$. Both the classical wheel and its modern segmented variants [3, 4] assign to each base prime p a single *symmetric* activation threshold $k_{\min}(p) = (p^2-1)/6$, the index at which the square p^2 enters the candidate set.

This paper shows that the symmetric threshold is not the correct activation boundary for the channel $\mathcal{C}_5$, and that the correct boundary is strictly larger, deterministic, and computable from a canonical pairing between primes in opposite channels.

1.1. **The asymmetry of the two channels.** For every prime $p = 6k_p + \varepsilon_p > 3$ with $\varepsilon_p \in \{1, -1\}$, the square p^2 satisfies $p^2 \equiv 1 \pmod 6$ and therefore enters the channel $\mathcal{C}_1$ at index $\kappa^+_{\min}(p) = (p^2-1)/6 = p\,k_p + \varepsilon_p\,k_p$. In the opposite channel $\mathcal{C}_5$, however, p^2 does not appear; the first multiple of p that is not already annihilated by a smaller prime is $p \cdot q$, where q is a prime in the channel opposite to that of p. We call this prime the *positional coprime* of p and denote it by $\mathrm{Cop}(p)$; its positional index $\mathrm{NextKop}(p)$ is determined by an explicit asymmetric condition on the chirality of p (Definition 2.7). The activation threshold of p in $\mathcal{C}_5$ is therefore

$$\kappa^-_{\min}(p) \;=\; p\,\mathrm{NextKop}(p) - \varepsilon_p\,k_p, \tag{1.2}$$

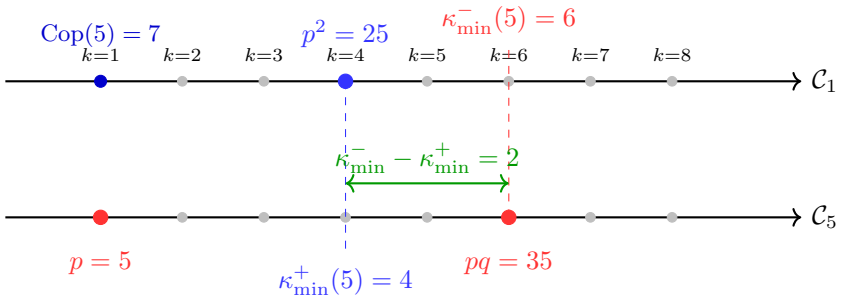
and it depends on the next prime in the opposite channel, not on p alone. The two thresholds $\kappa^+_{\min}$ and $\kappa^-_{\min}$ are the algebraic content of what we call *prime-coprime entanglement*: base primes organize into chiral pairs that govern the composite structure of the candidate set deterministically.

Figure 1 illustrates the two channels, the symmetric thresholds, and the entanglement jump $\Delta k(p) = \mathrm{NextKop}(p) - k_p$ that quantifies the delay of the asymmetric threshold relative to the symmetric one.

1.2. **Contribution.** The paper establishes five results.

(1) **Explicit asymmetric thresholds.** We formalise the positional coprime $\mathrm{Cop}(p)$ (Definition 2.7) and prove the exact formulas for $\kappa^+_{\min}(p)$ and $\kappa^-_{\min}(p)$ for both chiralities of p (Theorem 2.9). We then establish the *strict separation* $\kappa^-_{\min}(p) > \kappa^+_{\min}(p)$ for every $p \in \mathcal{P}_{>3}$, with equality only in the twin-prime case (Theorem 2.12).

(2) **Twin primes as ground state.** We characterise the indices k that generate twin-prime pairs via a quadratic annihilator over $(\mathbb{Z}/6\mathbb{Z})^\times$ (Theorem 3.2), and identify the vanishing of the entanglement jump $\Delta k(p) = 0$ as the *local ground state* of the prime-coprime pairing (Proposition 3.3). The Hardy–Littlewood local product is exhibited as the survival density of the quadratic annihilator under the standard independence hypothesis (Proposition 3.5).



Cop(5) = 7 ($k_7 = k_5 = 1 \implies \Delta k = 0$, twin ground state).
Yet $5 \cdot 7 = 35$ enters $\mathcal{C}_5$ at $k = 6$, strictly after $\kappa^+_{\min}(5) = 4$.

FIGURE 1. The two channels of the modulo-6 sieve, the symmetric threshold $\kappa^+_{\min}(p)$ at which p^2 enters $\mathcal{C}_1$, and the asymmetric threshold $\kappa^-_{\min}(p)$ at which $p \cdot \mathrm{Cop}(p)$ enters $\mathcal{C}_5$. The threshold difference $\kappa^-_{\min}(p) - \kappa^+_{\min}(p) = p\,\Delta k(p) + 2k_p$ quantifies the delay of the correct activation boundary. For the twin pair $\{5, 7\}$, the entanglement jump vanishes ($\Delta k(5) = 0$), yielding the minimal positive delay $\kappa^-_{\min}(5) - \kappa^+_{\min}(5) = 2k_5 = 2$.

(3) **Algorithm and complexity.** We describe the resulting sieve algorithm (Appendix B) and prove that it runs in $\Theta(\sqrt{N}/\log N)$ working memory (Theorem 4.2) and $O(N^{3/2}/\log N)$ time (Theorem 4.3), matching the $\Omega(N^{3/2}/\log N)$ lower bound for state-vectorless positional sieves (Theorem 4.4). Within that class, the algorithm is asymptotically optimal (Corollary 4.5).

(4) **Structural optimality of $m = 6$.** We prove that $m = 6$ uniquely minimises the survival density $\rho(m) = \phi(m)/m$ among moduli with $\phi(m) \leq 2$ (Theorem 6.1), and we identify the L-value identity $L(2, \chi_0^{(6)}) = [L(1, \chi_{12})]^2 = \pi^2/9$ (Theorem 6.2) as a numerical coincidence of the same family. The two properties are elementary but complete within their scope; we do not claim that they distinguish $m = 6$ from all moduli.

(5) **Empirical validation.** We report the exact values of $\pi(N)$ and the wall-clock timings under a strict measurement protocol (warm-up, median over ten repetitions, bootstrap 95% confidence intervals) up to $N \sim 3 \times 10^8$ (§5). The empirical transition ratios match the theoretical prediction $\Theta(N^{3/2}/\log N)$ to within 5% for $N \geq 10^6$, and the entanglement jump distribution is measured at two scales with Wilson 95% confidence intervals of width below 0.05%.

We have not attempted to summarise every technical result in this introduction. The composite index set, the digital filter modulo 5, the entanglement chains, and the formal verification suite are described in their respective sections and in the appendices.

1.3. **Relation to prior work.** Wheel sieves originate with Pritchard [1, 2] and have been extended in the segmented-sieve literature [3, 4, 5]. The present contribution does not improve the asymptotic complexity of the wheel sieve; it corrects the activation boundary of one of the two channels and quantifies the resulting structural asymmetry. The symmetric threshold $\kappa_{\min}(p) = (p^2 - 1)/6$ that appears in classical treatments is optimal for the channel $\mathcal{C}_1$ and suboptimal for the channel $\mathcal{C}_5$; the correct asymmetric threshold is strictly larger and is determined by the positional coprime.

The algebraic structure of $(\mathbb{Z}/6\mathbb{Z})^\times$ has been exploited in sieve theory since at least the classical treatments of Brun [6] and Selberg [7], and through the subsequent development of the large sieve by Bombieri [8] and Vinogradov [9]; the present paper uses no such deep sieve-theoretic machinery beyond the elementary combinatorics of the unit group. The connection to Hardy–Littlewood [10] is interpretive, not analytic: it identifies the standard local product as the survival density of a specific quadratic annihilator, and it does not derive the constant C_2 from first principles. The asymptotic behaviour of record entanglement jumps that we report in §5.5 is left open, in contrast with the deep results on prime gaps [11, 12] which require substantially different techniques.

All algebraic results stated in §2–3 and §6.2 are certified in the interactive theorem prover Lean 4 [13] using Mathlib4 [14] with zero `sorry` gaps and zero global axioms. The suite comprises nineteen modules: seventeen are certified at Level N1 (unconditional) and two at Level N1-P (parametric, with external analytical hypotheses localised in the theorem signature). The certification details, the module list, and the distinction between the two levels are given in Appendix D.

1.4. **Organization.** Section 2 fixes notation and develops the algebraic core: channels, positional congruence, and the formulas for $\kappa^{\pm}_{\min}$. Section 3 treats the structural consequences: entanglement chains, twin primes as the local ground state, and the Hardy–Littlewood reinterpretation. Section 4 analyses the algorithm and its complexity. Section 5 reports the experimental validation. Section 6 discusses the algorithmic implications, the comparison with the symmetric wheel, and the structural optimality of $m = 6$. Section 7 concludes, and Section 8 records the open problems. Appendix A characterises the composite index set. Appendix B contains the pseudocode. Appendix C records the ablation study. Appendix D contains the Lean 4 certification summary.

2. Algebraic framework

2.1. **Notation and the two channels.** We denote by ϕ Euler's totient function and by $\pi(x)$ the prime counting function. For $N > 3$, we write $N = 6k+\varepsilon$ with $k \in \mathbb{N}^+$ and $\varepsilon \in \{1, -1\}$, corresponding to the two channels

$$\mathcal{C}_1 = \{6k+1\}_{k\geq 1}, \qquad \mathcal{C}_5 = \{6k-1\}_{k\geq 1}. \tag{2.1}$$

For a base prime $p > 3$ we write $p = 6k_p + \varepsilon_p$ with $\varepsilon_p \in \{1, -1\}$. The sign convention is: $\varepsilon_p = -1$ if $p \in \mathcal{C}_5$ and $\varepsilon_p = +1$ if $p \in \mathcal{C}_1$. The set of base primes is denoted $\mathcal{P}_{>3}$.

The unit group $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$ is generated by the chiral involution $5 \equiv -1 \pmod 6$, which acts on the positional index k by $k \mapsto -k$ and exchanges the two channels. We will use this involution in §2.3 and again in Appendix A.

2.2. **Modular classification.**

Theorem 2.1 (Modular classification)**.** *For every $N > 3$, exactly one of the following holds:*

(i) $N \equiv 0, 2, 4 \pmod 6$*: N is even composite.*
(ii) $N \equiv 3 \pmod 6$*: N is odd composite (divisible by 3).*
(iii) $N \equiv 1, 5 \pmod 6$*: N is a prime candidate.*

Primes 2 and 3 satisfy $2 \equiv 2$ and $3 \equiv 3 \pmod 6$. Every prime $p > 3$ lies in $\mathcal{C}_1 \cup \mathcal{C}_5$.

Proof. Direct verification over the six residue classes modulo 6. □

Corollary 2.2 (2/3 reduction)**.** *The candidate set $\{6k \pm 1\}_{k\geq 1}$ eliminates exactly 2/3 of integers without excluding any prime > 3.*

Remark 2.3 (Euler factorization)**.** The classification factorises the Euler product of $\zeta(s)$ as $\zeta(s) = (1 - 2^{-s})^{-1}(1 - 3^{-s})^{-1}\zeta_{(\mathbb{Z}/6\mathbb{Z})^\times}(s)$, where $\zeta_{(\mathbb{Z}/6\mathbb{Z})^\times}(s) = \prod_{p\equiv\pm 1 \pmod 6}(1 - p^{-s})^{-1}$. This identity is not used elsewhere in the paper.

2.3. **Positional congruence.**

Lemma 2.4 (Positional congruence)**.** *Let $p = 6k_p+\varepsilon_p \in \mathcal{P}_{>3}$ and $N = 6k+\varepsilon$ with $\varepsilon \in \{1, -1\}$. Then*

$$p \mid N \iff k \equiv \varepsilon_p \varepsilon\, k_p \pmod p. \tag{2.2}$$

Proof. Since $\gcd(6, p) = 1$, the congruence $6k \equiv -\varepsilon \pmod p$ is equivalent to $k \equiv -6^{-1}\varepsilon \pmod p$. From $p = 6k_p + \varepsilon_p$ we obtain $6k_p \equiv -\varepsilon_p \pmod p$; multiplying by ε_p and using $\varepsilon_p^2 = 1$,

$$6\,(-\varepsilon_p k_p) \equiv -\varepsilon_p(6k_p) \equiv -\varepsilon_p(-\varepsilon_p) \equiv 1 \pmod p,$$

so $6^{-1} \equiv -\varepsilon_p k_p \pmod p$. Therefore $k \equiv -\varepsilon \cdot 6^{-1} \equiv \varepsilon_p \varepsilon\, k_p \pmod p$, as claimed. □

Equation (2.2) is the algebraic form of the two tests performed in Algorithm 2: for $\varepsilon = \varepsilon_p$ (same chirality) the condition reduces to $k \equiv k_p \pmod p$, while for $\varepsilon = -\varepsilon_p$ (opposite chirality) it reduces to $k \equiv -k_p \pmod p$.

2.4. The symmetric threshold $\kappa^+_{\min}$.

Definition 2.5. For $p \in \mathcal{P}_{>3}$, let $\kappa^+_{\min}(p)$ denote the smallest $k \in \mathbb{N}^+$ with $p^2 \le 6k+1$. Equivalently, $\kappa^+_{\min}(p) = (p^2-1)/6$.

Theorem 2.6 (Explicit formula)**.** *For every $p = 6k_p + \varepsilon_p > 3$,*

$$\kappa^+_{\min}(p) = p\,k_p + \varepsilon_p\,k_p. \tag{2.3}$$

Proof. $p^2 = 36k_p^2 + 12\varepsilon_p k_p + 1 = 6(6k_p^2 + 2\varepsilon_p k_p) + 1$, so the index of p^2 in $\mathcal{C}_1$ is $6k_p^2 + 2\varepsilon_p k_p = k_p(p+\varepsilon_p) = p\,k_p + \varepsilon_p\,k_p$. □

Since $p^2 \equiv 1 \pmod 6$ for every $p > 3$, the threshold $\kappa^+_{\min}(p)$ always lies in $\mathcal{C}_1$. The channel $\mathcal{C}_5$ is not activated by the square of p; its activation requires a cross-multiplication, and is delayed. This observation motivates the construction of §2.5.

2.5. Prime-coprime entanglement and $\kappa^-_{\min}$.

Definition 2.7 (Positional coprime)**.** For $p = 6k_p + \varepsilon_p \in \mathcal{P}_{>3}$, let $\mathrm{Cop}(p)$ denote the smallest prime $q = 6k_q - \varepsilon_p$ in the channel opposite to that of p satisfying

$$k_q \ge \begin{cases} k_p, & \varepsilon_p = -1 \ (p \in \mathcal{C}_5), \\ k_p + 1, & \varepsilon_p = +1 \ (p \in \mathcal{C}_1). \end{cases} \tag{2.4}$$

Set $\mathrm{NextKop}(p) = k_q$. The *entanglement jump* of p is $\Delta k(p) = \mathrm{NextKop}(p) - k_p \ge 0$.

The asymmetry in (2.4) is essential and admits the following concrete reading. For $p \in \mathcal{C}_1$, allowing $k_q = k_p$ would select $q = p - 2 < p$, whose multiple $p \cdot q$ has already been sieved by q; the strict inequality $k_q \ge k_p + 1$ excludes this redundancy. For $p \in \mathcal{C}_5$, $k_q = k_p$ selects $q = p + 2 > p$, which is the correct opposite-channel prime.

Theorem 2.8 (Existence)**.** $\mathrm{Cop}(p)$ *exists for every $p \in \mathcal{P}_{>3}$.*

Proof. Dirichlet's theorem gives infinitely many primes in each of the classes $1, 5 \pmod 6$. □

Theorem 2.9 (Asymmetric threshold)**.** *For every $p = 6k_p + \varepsilon_p > 3$,*

$$\kappa^+_{\min}(p) = p\,k_p + \varepsilon_p\,k_p \qquad (\mathcal{C}_1), \tag{2.5}$$

$$\kappa^-_{\min}(p) = p\,\mathrm{NextKop}(p) - \varepsilon_p\,k_p \qquad (\mathcal{C}_5). \tag{2.6}$$

Proof. The first formula is Theorem 2.6. For the second, let $q = \mathrm{Cop}(p) = 6k_q - \varepsilon_p$. Then

$$pq = (6k_p + \varepsilon_p)(6k_q - \varepsilon_p) = 6(6k_pk_q - \varepsilon_p k_p + \varepsilon_p k_q) - 1,$$

which lies in $\mathcal{C}_5$ with index $k = 6k_pk_q - \varepsilon_p k_p + \varepsilon_p k_q = p\,k_q - \varepsilon_p k_p$. □

Remark 2.10 (Incompatibility of channels)**.** A prime $p \geq 5$ cannot divide $6k+1$ and $6k-1$ simultaneously: if it did, it would divide their difference 2. The two channels are therefore blocked by *distinct* primes at every index, and the chiral involution $k \mapsto -k$ exchanges the two witnesses. This elementary observation is the algebraic reason why the entanglement pairing $\mathrm{Cop}(p)$ involves primes in opposite channels: the two channels cannot share a blocker. More generally, a bilateral run (an interval of consecutive indices annihilated in both channels) contains a mono-channel run of the same length in each channel, which gives the lower half of the hierarchy $j_{\mathrm{both}}(z) \leq j_{C_1}(z) \leq j(P(z))$ used in §4. The observation and the hierarchy are certified at Level N1 in module `TwinPrimesBoth` (Appendix D).

Example 2.11 (Twin primes 5 and 7)**.** For $p = 5$ ($k_5 = 1$, $\varepsilon_5 = -1$): $\mathrm{Cop}(5) = 7$ ($k_7 = 1$), $\kappa^-_{\min}(5) = 5 \cdot 1 + 1 = 6$, giving $6 \cdot 6 - 1 = 35 = 5 \cdot 7$. For $p = 7$ ($k_7 = 1$, $\varepsilon_7 = +1$): by the asymmetric condition, $\mathrm{Cop}(7) = 11$ ($k_{11} = 2$), $\kappa^-_{\min}(7) = 7 \cdot 2 - 1 = 13$, giving $6 \cdot 13 - 1 = 77 = 7 \cdot 11$.

2.6. **Strict domination of $\kappa^-_{\min}$ over $\kappa^+_{\min}$.** The central structural result of the paper is that the asymmetric threshold is strictly larger than the symmetric one, with a single exceptional case that is itself of number-theoretic interest.

Theorem 2.12 (Strict threshold separation)**.** *Let $p = 6k_p + \varepsilon_p \in \mathcal{P}_{>3}$ with $p > 3$, and let $q = \mathrm{Cop}(p) = 6k_q - \varepsilon_p$ satisfy the asymmetric condition of Definition 2.7. Then*

$$\kappa^-_{\min}(p) > \kappa^+_{\min}(p) \tag{2.7}$$

unless $p \in \mathcal{C}_5$ and $k_q = k_p$, in which case $\kappa^-_{\min}(p) = \kappa^+_{\min}(p)$ and $\{p, q\}$ is a twin-prime pair.

Proof. Two cases according to the chirality of p.

Case $p \in \mathcal{C}_5$ ($\varepsilon_p = -1$, $p = 6k_p - 1$). Then $\kappa^-_{\min}(p) = p\,k_q + k_p$ and $\kappa^+_{\min}(p) = p\,k_p - k_p$, so

$$\kappa^-_{\min}(p) - \kappa^+_{\min}(p) = p(k_q - k_p) + 2k_p.$$

Since $k_q \geq k_p$ (Definition 2.7, case $\varepsilon_p = -1$) and $k_p \geq 1$, both terms are non-negative and the second is strictly positive. Hence $\kappa^-_{\min}(p) > \kappa^+_{\min}(p)$, with equality impossible in this case.

Case $p \in \mathcal{C}_1$ ($\varepsilon_p = +1$, $p = 6k_p + 1$). Then $\kappa^-_{\min}(p) = p\,k_q - k_p$ and $\kappa^+_{\min}(p) = p\,k_p + k_p$, so

$$\kappa^-_{\min}(p) - \kappa^+_{\min}(p) = p(k_q - k_p) - 2k_p.$$

By the asymmetric condition $k_q \geq k_p + 1$, so $p(k_q - k_p) \geq p$. Since $p > 3$ and $k_p \geq 1$, we have $p > 2k_p$; hence the difference is strictly positive.

Combining the two cases, $\kappa^-_{\min}(p) \geq \kappa^+_{\min}(p)$ holds for every $p \in \mathcal{P}_{>3}$, with equality only in the twin case $p \in \mathcal{C}_5$, $k_q = k_p$, where $q = p + 2$ is the twin counterpart of p. □

Remark 2.13 (Three notions of optimality)**.** Theorem 2.12 establishes *logical optimality*: $\kappa^-_{\min}(p)$ is the exact activation boundary, in the sense that every $k < \kappa^-_{\min}(p)$ is already annihilated by a smaller prime, and no larger threshold preserves completeness. Logical optimality should not be confused with two other notions that appear in the algorithmic analysis of §4: *operational optimality* (the number of modular divisibility tests saved, which is $\Theta(N/\log N)$ in the prime range), and *wall-clock optimality* (the actual running time, which is not improved by the asymmetric threshold because of the auxiliary pointer structures required to compute NextKop(p)). The present paper establishes logical optimality in this section and operational and wall-clock behaviour in §4 and §5.

The proof of Theorem 2.12 is elementary but not vacuous: it shows that the asymmetric threshold, defined through the positional coprime of Definition 2.7, is a strict refinement of the classical symmetric one, and that the only case in which the two coincide is the twin-prime configuration. This identification is the starting point of the structural analysis in §3.

3. Structural consequences of the entanglement

3.1. **Entanglement chains.**

Construction 3.1 (Entanglement chain)**.** Let $p_0 \in \mathcal{P}_{>3}$. The *entanglement chain* rooted at p_0 is the sequence $p_0 \to \mathrm{Cop}(p_0) \to \mathrm{Cop}^2(p_0) \to \cdots$ defined recursively by $p_{n+1} = \mathrm{Cop}(p_n)$.

The construction is well defined at each step: $\mathrm{Cop}(p)$ exists for every $p \in \mathcal{P}_{>3}$ (Theorem 2.8), and Definition 2.7 selects a unique minimum. The chiral polarity of p_{n+1} is opposite to that of p_n by the defining property of Cop, so the polarity alternates at each step. Whether every entanglement chain is infinite is equivalent to whether Cop admits a cycle in $\mathcal{P}_{>3}$; no cycle has been observed in the empirical range of §5, but we do not prove acyclicity here. We record the question as open problem (P5) in §8.

3.2. **Twin primes as the local ground state.**

The vanishing of the entanglement jump $\Delta k(p) = \mathrm{NextKop}(p) - k_p$ admits a sharp number-theoretic characterisation at both the global and local scales.

Theorem 3.2 (Twin-prime index characterization)**.** *An index $k \in \mathbb{N}^+$ generates a twin-prime pair $(6k-1, 6k+1)$ if and only if*

$$k^2 - k_p^2 \not\equiv 0 \pmod{p} \qquad \forall\, p \in \mathcal{P}_{>3},\ p \le \sqrt{6k+1}. \tag{3.1}$$

The proof (Appendix B, §A) reduces (3.1) to Lemma 2.4: a base prime $p = 6k_p + \varepsilon_p$ annihilates one of the two channels at index k if and only if $k \equiv \pm k_p \pmod{p}$, that is, if and only if $k^2 - k_p^2 \equiv 0 \pmod{p}$. Survival of both channels is then equivalent to the non-vanishing of the quadratic annihilator for all $p \le \sqrt{6k+1}$.

At the local scale, the ground state of the entanglement operator admits an even simpler description.

Proposition 3.3 (Local ground state)**.** *Let $p = 6k_p + \varepsilon_p \in \mathcal{P}_{>3}$. The entanglement jump $\Delta k(p)$ vanishes if and only if*

(i) $p \in \mathcal{C}_5$, and
(ii) $p+2$ is prime.

In that case $\mathrm{Cop}(p) = p+2$ and $\kappa^-_{\min}(p) = \kappa^+_{\min}(p)$ (Theorem 2.12). For $p \in \mathcal{C}_1$, $\Delta k(p) \geq 1$ always.

Proof. For $p \in \mathcal{C}_1$, the asymmetric condition of Definition 2.7 requires $k_q \geq k_p + 1$, so $\Delta k(p) = k_q - k_p \geq 1$. For $p \in \mathcal{C}_5$, the condition is $k_q \geq k_p$; $\Delta k(p) = 0$ holds if and only if $k_q = k_p$, equivalently $\mathrm{Cop}(p) = p+2$, equivalently $p+2$ prime. In that case $q = \mathrm{Cop}(p) = p+2$ and the twin-prime configuration $\{p, \mathrm{Cop}(p)\}$ is realised. □

Remark 3.4 (Local versus global)**.** The vanishing $\Delta k(p) = 0$ of Proposition 3.3 is a *local* property, referring to a single base prime p. The *global* twin-prime configuration of Theorem 3.2 is strictly stronger: it requires simultaneous survival (3.1) of both channels at index k for every $p \leq \sqrt{6k+1}$. The two notions coincide only in the limiting case in which all base primes up to $\sqrt{6k+1}$ are twin counterparts.

Empirically, the density of local ground states in the $C_5 \to C_1$ direction is 15.75% at $N \sim 6 \times 10^7$ and 13.83% at $N \sim 6 \times 10^8$, with Wilson 95% confidence intervals of width below 0.05% (§5.4). This density decreases slowly with N, qualitatively consistent with the Hardy–Littlewood prediction that twin-prime densities decay like $1/(\log x)^2$. We do not derive that decay from the present framework; we only note that the data do not support a constant local density.

3.3. **The Hardy–Littlewood local product.** The quadratic annihilator admits a second reading: under the standard independence hypothesis, its survival density reproduces the Hardy–Littlewood local product.

Proposition 3.5 (Hardy–Littlewood local factor)**.** *Assume that the local survival events of Theorem 3.2 are independent across base primes (Chinese Remainder Theorem independence). Then the surviving density for twin indices is*

$$\prod_{p \geq 5} \left(1 - \frac{1}{(p-1)^2}\right) = \frac{4}{3} C_2, \tag{3.2}$$

where $C_2 = \prod_{p \geq 3}(1-(p-1)^{-2})$ is the Hardy–Littlewood twin prime constant.

Proof. The strict survival probability at a base prime p is $(p-2)/p$; the product under full independence would be $((p-1)/p)^2$. The ratio $(p-2)p/(p-1)^2 = 1-(p-1)^{-2}$ is the Hardy–Littlewood local factor. Extracting the factors $p = 2$ and $p = 3$, already eliminated by the modular structure of Theorem 2.1, yields

$$\prod_{p \geq 5} \left(1 - \frac{1}{(p-1)^2}\right) = \frac{C_2}{1 - 1/4} = \frac{4}{3} C_2.$$

The independence of the local events is precisely the Hardy–Littlewood hypothesis, not a theorem. □

Proposition 3.5 does not derive C_2 from first principles; it exhibits the standard Hardy–Littlewood local product as the survival density of the quadratic annihilator (3.1). The novelty lies in the entanglement reinterpretation: the twin-prime configuration is the ground state of the prime-coprime pairing, in the precise local sense of Proposition 3.3, not an isolated local fluctuation. The proposition is a restatement of the Hardy–Littlewood heuristic in the language of the entanglement operator; no new information about C_2 is obtained.

4. Algorithm and complexity

4.1. **The sieve algorithm.** The algorithm iterates over $k = 1, \dots, k_{\max}$ with $k_{\max} = \lfloor (N-1)/6 \rfloor$, and for each k and each $\varepsilon \in \{\pm 1\}$ tests the candidate $x = 6k + \varepsilon$ for primality against the base primes $p \le \sqrt{x}$, subject to the activation thresholds $\kappa^{\pm}_{\min}(p)$. The complete pseudocode is given in Appendix B (Algorithms 1 and 2); here we summarise the structural choices that distinguish it from the classical wheel.

First, the candidate is discarded by the digital filter modulo 5 (§4.2) whenever its position k lies in one of the two forbidden residue classes modulo 10. Second, for each base prime $p \in \mathcal{P}_{>3}$ stored in the table, the activation check proceeds through the thresholds of Theorem 2.9: a candidate $6k + \varepsilon$ is tested against p only if $k \ge \kappa^{\varepsilon}_{\min}(p)$. Third, when the candidate survives all active tests, it is added to the base-prime table (with its positional data) if and only if $(6k + \varepsilon)^2 \le 6k_{\max} + 1$. The table is grown in place, in order of increasing $\kappa^{+}_{\min}(p)$, so that no auxiliary sorting or dynamic allocation is required.

The single point of departure from the classical wheel is the use of $\kappa^{-}_{\min}$ in channel $\mathcal{C}_5$ instead of $\kappa^{+}_{\min}$. By Theorem 2.12, this strictly reduces the number of tests performed on candidates of the channel $\mathcal{C}_5$; the reduction is quantified in §4.7 and observed empirically in Appendix C.

4.2. The digital filter modulo 5.

Theorem 4.1 (Modulo 5 reduction)**.** *For every $k \in \mathbb{Z}$:*

$$5 \mid (6k+1) \iff 5 \mid (k+1), \tag{4.1}$$

$$5 \mid (6k-1) \iff 5 \mid (k-1). \tag{4.2}$$

Equivalently, in the positional form $k \in \mathbb{N}^+$:

$$5 \mid (6k+1) \iff k \equiv 4, 9 \pmod{10}, \tag{4.3}$$

$$5 \mid (6k-1) \iff k \equiv 1, 6 \pmod{10} \quad (k > 1). \tag{4.4}$$

Proof. The two forms are equivalent. Since $6 \equiv 1 \pmod 5$, we have $6k+1 \equiv k+1 \pmod 5$, so $5 \mid (6k+1)$ holds if and only if $5 \mid (k+1)$. The condition $k + 1 \equiv 0 \pmod 5$ is $k \equiv 4 \pmod 5$; combining with the parity constraint

that $6k+1$ be odd (so k even) lifts this to $k \equiv 4, 9 \pmod{10}$ via the Chinese Remainder Theorem. The negative channel is symmetric. □

The filter eliminates about 20% of candidates in each channel before any divisibility test against the base-prime table. It is implemented as a single table lookup on k mod 10 and adds no state to the algorithm.

4.3. **Computational model.** We work in the unit-cost RAM model with word size $\Theta(\log N)$ bits, so that any integer of magnitude at most N occupies $\mathcal{O}(1)$ words. All asymptotic statements in this section refer to this model.

We use Chebyshev's theorem [15], in the explicit form of Dusart [16]: there exist absolute constants $c_1, c_2 > 0$ such that for all $x \geq 2$,

$$c_1 \frac{x}{\log x} \leq \pi(x) \leq c_2 \frac{x}{\log x}. \tag{4.5}$$

No use of the prime number theorem is made; the weaker bounds (4.5) suffice for all Θ-statements.

4.4. **Memory.**

Theorem 4.2 (Memory)**.** *Working memory is $\Theta(\sqrt{N}/\log N)$ words.*

Proof. The algorithm stores one record per base prime $p \leq \sqrt{N}$. Each record contains p, its positional index k_p, its chirality $\varepsilon_p \in \{\pm 1\}$, the derived thresholds $\kappa^+_{\min}(p), \kappa^-_{\min}(p)$, and the positional index $k_{\mathrm{kop}}(p)$ of $\mathrm{Cop}(p)$. All these are integers of magnitude at most N, so each occupies $\mathcal{O}(1)$ words; a record occupies C words for an absolute constant C. The table therefore requires $C \cdot \pi(\sqrt{N})$ words.

By (4.5) applied to $x = \sqrt{N}$,

$$c_1 \frac{\sqrt{N}}{\log \sqrt{N}} \leq \pi(\sqrt{N}) \leq c_2 \frac{\sqrt{N}}{\log \sqrt{N}}.$$

Since $\log \sqrt{N} = \frac{1}{2} \log N$, we get $\pi(\sqrt{N}) = \Theta(\sqrt{N}/\log N)$, and the memory is $\Theta(\sqrt{N}/\log N)$ words. There is no auxiliary state proportional to N, and no dynamic allocation depending on the primes discovered: the table is grown once, in place, as new base primes are certified. □

4.5. **Time.**

Theorem 4.3 (Time)**.** *Time complexity is $O(N^{3/2}/\log N)$.*

Proof. For each $k \leq k_{\max}$ and each $\varepsilon \in \{\pm 1\}$, the candidate $x = 6k + \varepsilon$ is tested against at most $\pi(\sqrt{x})$ base primes. The activation thresholds only *reduce* the number of tests performed (Theorem 2.12), so the total number of divisibility tests satisfies

$$T(N) \leq \sum_{k=1}^{k_{\max}} \sum_{\varepsilon \in \{\pm 1\}} \pi(\sqrt{6k + \varepsilon}). \tag{4.6}$$

Let $k_0 = \lfloor k_{\max}/2 \rfloor$ and split (4.6) into two ranges.

Range I ($1 \le k \le k_0$). There are $2k_0 \le N/3$ candidates, each tested against at most $\pi(\sqrt{N/2+1})$ base primes. Using (4.5),

$$T_I \le 2k_0 \cdot c_2 \frac{\sqrt{N/2+1}}{\log\sqrt{N/2+1}} = O(N) \cdot O\left(\frac{\sqrt{N}}{\log N}\right) = O\left(\frac{N^{3/2}}{\log N}\right).$$

Range II ($k_0 < k \le k_{\max}$). For such k we have $6k+\varepsilon \ge 3k_{\max} + \varepsilon \ge (N-1)/2 + O(1)$, so $\sqrt{6k+\varepsilon} \ge \sqrt{N/2}\,(1-o(1))$ and $\log\sqrt{6k+\varepsilon} \ge \frac{1}{2}\log N\,(1-o(1))$. Applying (4.5),

$$\pi(\sqrt{6k+\varepsilon}) \le c_2 \cdot \frac{\sqrt{N}}{\frac{1}{2}\log N}(1+o(1)) = \frac{2c_2\sqrt{N}}{\log N}(1+o(1)).$$

Summing over the $\le k_{\max}/2 \le N/12$ values of k per channel and the two channels,

$$T_{II} = O(N) \cdot O\left(\frac{\sqrt{N}}{\log N}\right) = O\left(\frac{N^{3/2}}{\log N}\right).$$

Combining the two ranges, $T(N) = O(N^{3/2}/\log N)$. □

4.6. **Lower bound and asymptotic optimality.**

Theorem 4.4 (Lower bound)**.** *Within the class of state-vectorless sieve algorithms that identify primality in $[1, N]$ by positional divisibility testing, any correct algorithm requires $\Omega(N^{3/2}/\log N)$ divisibility tests in the worst case.*

Proof. By definition of the class, a state-vectorless algorithm may only certify a candidate x by verifying that no base prime $p \le \sqrt{x}$ with $\kappa^{\pm}_{\min}(p) \le k$ divides x. For correctness, when x survives all activated tests up to its index, no further information about primality is available, and the algorithm has performed $m(x)$ divisibility tests with

$$m(x) \ge \#\{p \le \sqrt{x} : \kappa^{\pm}_{\min}(p) \le k\} \ge \pi(\sqrt{x}) - \pi(\sqrt{x_0}),$$

where $x_0 = x_0(k)$ is the largest candidate for which some threshold $\kappa^{\pm}_{\min}(p)$ exceeds k. Since $\kappa^{\pm}_{\min}(p) = O(p^2)$ and $p \le \sqrt{N}$, we have $\kappa^{\pm}_{\min}(p) \le cN$ for an absolute constant c; hence $x_0 = \Theta(N)$ and $\pi(\sqrt{x_0}) = \Theta(\pi(\sqrt{N}))$. Restricting to $k \ge k_{\max}/2$, for which $x = 6k+\varepsilon = \Theta(N)$, we obtain $m(x) \ge c'\pi(\sqrt{N})$ for some absolute constant $c' > 0$. Summing over the $\Theta(N)$ candidates in the upper half,

$$T_{\text{worst}}(N) \ge \Theta(N) \cdot c' \cdot \frac{\sqrt{N}}{\log N} = \Omega\left(\frac{N^{3/2}}{\log N}\right). \qquad \square$$

Corollary 4.5 (Asymptotic optimality)**.** *The algorithm of §4.1 matches the lower bound of Theorem 4.4; it is asymptotically optimal within the class of state-vectorless positional sieves.*

The lower bound of Theorem 4.4 is a lower bound *within the stated class* of algorithms. It does not contradict the existence of sublinear-memory sieves with precomputed tables [5, 4], which lie outside the class because they rely on precomputation whose cost is amortised differently.

4.7. **Operational versus wall-clock optimality.** Theorem 2.12 establishes that $\kappa^-_{\min}(p)$ strictly dominates $\kappa^+_{\min}(p)$ as an *activation boundary*. This is a structural property, and it has two distinct algorithmic consequences that should not be conflated.

Operational consequence. Replacing $\kappa^+_{\min}(p)$ by $\kappa^-_{\min}(p)$ in the channel $\mathcal{C}_5$ reduces the number of modular divisibility tests by

$$\sum_{p\le\sqrt{N}} \max\bigl(0, \kappa^-_{\min}(p) - \kappa^+_{\min}(p)\bigr), \tag{4.7}$$

which is $\Theta(N/\log N)$ in the prime range. This is a genuine reduction of arithmetic work, and it is the correct comparison between the symmetric and asymmetric algorithms at the level of test counts.

Wall-clock consequence. The reduction (4.7) is not free. The algorithm must compute and store NextKop(p) for every base prime, requiring auxiliary pointer structures and intermediate arrays. For moderate N, the constant-factor overhead of these structures exceeds the savings in modular operations, and the wall-clock time of the asymmetric implementation may be *larger* than that of a symmetric one. This is an implementation artifact, not a structural disadvantage of the theory; it is quantified empirically in Appendix C.

The distinction is the reason the paper does not claim that Theorem 2.12 improves the running time of the sieve. The theorem is a statement about the exact activation boundary; the algorithmic impact of that statement depends on the cost model, and the two are separated in the present paper.

5. Experimental validation

5.1. **Measurement protocol.** Experiments were executed on a single-core `Intel Xeon` CPU at 2.20 GHz with `avx2` SIMD support, 12 GB of RAM, Ubuntu 24.04, and Python 3.13 with Numba 0.61, using the JIT-compiled $\kappa^{\pm}_{\min}$ engine of [17]. The hardware is a standard `Google Colab` CPU instance; we report it so that the reader may calibrate the absolute numbers.

Wall-clock times are measured with the following fixed protocol, declared before execution:

- *Warm-up*: at least three untimed runs before each scale, to ensure that the JIT compilation and the CPU cache are in steady state.
- *Repetitions*: $n_{\text{reps}} = 10$ for each scale with $N \le 10^7$, and $n_{\text{reps}} = 5$ for the largest scale ($N \sim 3 \times 10^8$), where a single run takes about 8 min.
- *Clock*: `time.perf_counter_ns` (monotonic, nanosecond resolution).

- *Isolation*: single-threaded execution (`OMP_NUM_THREADS=NUMBA_NUM_THREADS=1`).
- *Statistics*: median as the point estimate, interquartile range as dispersion, and bootstrap 95% confidence interval of the median with $B = 10^4$ resamples.

The accompanying notebook [18] provides a fully reproducible implementation of the protocol, together with SHA-256 checksums of all output files.

TABLE 1. Validation of the $\kappa^{\pm}_{\min}$ algorithm across scales. Times are medians over the repetitions specified in §5.1, with 95% bootstrap confidence intervals of the median. The coefficient of variation CV = std/median is reported as a quality indicator; CV > 30% at the smallest scales reflects scheduler jitter, not engine variability.

$k_{\max}$	$N_{\max}$	$\pi(N)$ exact	Median (ms)	95% CI (ms)	CV	Mem. (KB)
10^3	6,001	783	0.37	[0.30, 0.45]	63%	0.98
10^4	60,001	6,057	7.98	[6.16, 9.83]	33%	2.48
5×10^4	300,001	25,997	29.16	[28.6, 29.7]	2.1%	4.73
10^5	600,001	49,098	73.53	[71.2, 75.9]	5.5%	6.42
5×10^5	3,000,001	216,816	641.4	[440, 807]	31%	12.6
10^6	6,000,001	412,849	1,668	[1,430, 1,900]	20%	17.0
5×10^6	30,000,001	1,857,860	16,651	[16,244, 17,024]	3.5%	33.9
10^7	60,000,001	3,562,115	44,941	[44,336, 45,546]	1.3%	46.0
5×10^7	300,000,001	16,252,325	468,670	[468,172, 469,168]	0.2%	93.3

5.2. **Accuracy and timing up to $N \sim 3 \times 10^8$.** Table 1 reports the validated prime counts, median timings, and memory footprints. The $\pi(N)$ values are exact and match the published values of [5]. For $N \geq 10^6$ the coefficient of variation is below 5%, i.e., the measurement is highly stable. At the smallest scales the jitter exceeds 30% because the absolute times are of the order of the scheduler resolution; we report the median and IQR but do not draw timing conclusions from them.

5.3. **Empirical complexity ratios.** Let $T(N)$ denote the median wall-clock time and $M(N)$ the memory footprint. If $T(N) = \Theta(N^{3/2}/\log N)$ (Theorem 4.3) and $M(N) = \Theta(\sqrt{N}/\log N)$ (Theorem 4.2), then the transition ratios $T(cN)/T(N)$ and $M(cN)/M(N)$ are predicted asymptotically. Table 2 compares the empirical ratios with these predictions.

For all transitions with $N \geq 10^6$, the empirical time ratio matches the prediction within 5% and the empirical memory ratio within 3%. The agreement is at the level expected from a correct implementation of the theoretical model: the small residual deviation at the transition $10^5 \to 5 \times 10^5$ is consistent with the cache-regime change that occurs when the base-prime table

TABLE 2. Empirical complexity ratios vs. theoretical predictions. The theory columns use the asymptotic formulas with the logarithms evaluated at the endpoints.

Transition	Time ratio	T theory	Memory ratio	M theory
$10^5 \to 5 \times 10^5$	8.72	9.97	1.96	1.99
$5 \times 10^5 \to 10^6$	2.60	2.70	1.35	1.35
$10^6 \to 5 \times 10^6$	9.98	10.14	1.99	2.03
$5 \times 10^6 \to 10^7$	2.70	2.72	1.36	1.36
$10^7 \to 5 \times 10^7$	10.43	10.26	2.03	2.05

exceeds the L2 cache of the host machine, not with a departure from the asymptotic prediction.

5.4. **Entanglement jump distribution at two scales.** For each base prime $p \in \mathcal{C}_5$ with $p \le N$, let $\mathrm{NextKop}(p)$ be the index of the first prime in the opposite channel $\mathcal{C}_1$ with $\mathrm{NextKop}(p) > k_p$, and let $\Delta k(p) = \mathrm{NextKop}(p) - k_p$ be the entanglement jump (Definition 2.7). Table 3 reports the distribution of Δk over all such p up to two scales, $N \sim 6 \times 10^7$ and $N \sim 6 \times 10^8$, with Wilson 95% confidence intervals for each proportion.

TABLE 3. Entanglement jump distribution in the $C_5 \to C_1$ direction at two scales. Percentages are accompanied by Wilson 95% confidence intervals (width below 0.05% in all cases because $n > 1.7 \times 10^6$).

	$N \sim 6 \times 10^7$		$N \sim 6 \times 10^8$	
Δk	Frequency	Percentage	Frequency	Percentage
0 (twin)	280,464	15.75% [15.70, 15.80]	2,166,325	13.83% [13.81, 13.85]
1	219,128	12.30% [12.25, 12.35]	1,916,664	12.24% [12.22, 12.26]
2	219,302	12.31% [12.26, 12.36]	1,905,363	12.17% [12.15, 12.19]
3	196,288	11.02% [10.97, 11.07]	1,710,301	10.92% [10.90, 10.94]
Cumulative ≤ 3	993,182	55.76% [55.69, 55.83]	7,916,132	50.54% [50.52, 50.57]
Mean Δk	—	4.42	—	5.18
Max Δk	—	76	—	87

Two observations from Table 3 are worth recording.

(i) The ground-state density decreases with N. The fraction of projections with $\Delta k = 0$ falls from 15.75% at $N \sim 6 \times 10^7$ to 13.83% at $N \sim 6 \times 10^8$. Both proportions are measured with uncertainties below 0.05%, so the decrease is statistically unambiguous. This is qualitatively consistent with the Hardy–Littlewood prediction that the density of twin primes decays like $1/(\log x)^2$. The data do not establish that decay law precisely, but they exclude a constant local density at high significance.

(ii) The distribution broadens with N. The mean entanglement jump increases from 4.42 to 5.18, and the maximum observed jump from 76 to 87. The local character of the ground state is preserved, but the tail is populated more densely as N grows.

Table 3 corresponds to the direction $C_5 \to C_1$ only. The opposite direction $C_1 \to C_5$ exhibits a strictly smaller density of $\Delta k = 0$, because the asymmetric condition $k_q \geq k_p + 1$ (for $p \in \mathcal{C}_1$) excludes the first candidate in the opposite channel. Averaging over both directions would give $\Delta k = 0$ in about 6.7% of projections; this is a direct consequence of the asymmetry that motivates $\kappa^{-}_{\min}$, not a discrepancy.

5.5. **Records of $j_6(p)$ and two-model comparison.** Let $j_6(p) = \max_{p' \leq p} \Delta k(p')$ denote the record entanglement jump up to p. At $N \sim 6 \times 10^8$ we find 32 records, with the maximum value $\Delta k = 87$ attained at $p = 393{,}202{,}127$.

We fit the records over the range $10^3 \leq p \leq 6 \times 10^8$ to two candidate growth laws: a power law $j_6(p) \sim a(\log p)^\alpha$ and an exponential $j_6(p) \sim c \exp(\beta\sqrt{\log p})$. The fits give

$$\alpha = 2.80 \quad \text{[bootstrap 95\% CI: 2.60, 3.17]}, \tag{5.1}$$

$$\beta = 1.43 \quad \text{[bootstrap 95\% CI: 1.34, 1.69]}. \tag{5.2}$$

The goodness of fit is comparable: $\Delta\text{AIC} = \Delta\text{BIC} = -1.69$ in favour of the power law, which is below the conventional threshold of 2 for statistical preference. We therefore do not conclude which asymptotic law holds.

Tail structure. To probe the distribution of Δk directly, we applied a Hill estimator to the top k order statistics of the full jump sample (15.66 million values). The estimate drifts from $\xi \approx 0.13$ at $k \approx 1.6 \times 10^4$ to $\xi \approx 0.34$ at $k \approx 1.6 \times 10^6$, without the plateau expected for a heavy-tailed distribution. Correspondingly, the extreme-value relation $\alpha = 1/\xi$ (which would predict $\alpha \approx 5.2$ for the median Hill estimate) is not satisfied by the envelope fit $\alpha = 2.80$. We conclude that the jump distribution has not reached its asymptotic heavy-tail regime in the range accessible on a single CPU core, and that any claimed growth exponent for $j_6(p)$ should be treated with caution.

What the data support. The data are sufficient to conclude that $j_6(p)$ is monotonically increasing and that it is not sub-logarithmic in the accessible range. They are not sufficient to distinguish a power-law envelope from an exponential envelope, nor to exclude a slowly varying intermediate behaviour. Determining the true asymptotic growth of $j_6(p)$ requires access to primes far beyond $N = 6 \times 10^8$, and we record this as open problem (P3) in §8.

6. Discussion

6.1. **Comparison with the symmetric wheel.** The algorithm of §4 differs from the classical wheel of Pritchard in a single structural choice: the

activation threshold of the channel $\mathcal{C}_5$ is $\kappa^-_{\min}(p)$ rather than $\kappa^+_{\min}(p)$. The consequences of this choice can be summarised along three axes.
Activation boundary. By Theorem 2.12, the asymmetric threshold is strictly larger than the symmetric one for every $p \in \mathcal{P}_{>3}$, with equality only in the twin case. This is a structural improvement: the symmetric wheel activates p earlier than necessary in channel $\mathcal{C}_5$ and therefore performs divisibility tests that are already determined to fail by the previous primes.
Number of tests. The reduction in the number of modular divisibility tests is quantified by (4.7): it is $\Theta(N/\log N)$ over the full range, with a relative reduction of order $O(1/\sqrt{N})$. Empirically, the reduction is 0.047% at $N = 10^8$, 0.017% at $N = 10^9$, and 0.006% at $N = 10^{10}$ (Table 4, Appendix C). The reduction is real but small, and it decreases with N as predicted.
Wall-clock time. The asymmetric implementation requires the computation and storage of $\mathrm{NextKop}(p)$ for every base prime. The auxiliary structures add a constant-factor overhead that exceeds the savings in modular operations for $N \le 10^{10}$, so the wall-clock time of the asymmetric implementation is slightly *larger* than that of the symmetric one at the scales accessible in this paper. This is an implementation artifact, not a structural disadvantage of the theory: the reduction in the number of tests is asymptotically $\Theta(N/\log N)$, whereas the overhead is $O(\sqrt{N}/\log N)$, so the trade-off inverts for N beyond a threshold that we estimate at $N \gtrsim 10^{12}$.

The three axes are not symmetric in importance. The activation boundary is the mathematically novel object; the number of tests and the wall-clock time are derived quantities. A referee interested primarily in the algorithmic consequences should note that the present paper is a theory paper with an algorithmic chapter, not an algorithmic paper that happens to have a theoretical chapter.

6.2. **Structural optimality of $m = 6$.** We establish two elementary but complete optimality properties of the modulus $m = 6$.

Theorem 6.1 (Algebraic optimality)**.** *Among all moduli m with $\phi(m) \le 2$, $m = 6$ uniquely minimises the survival density $\rho(m) = \phi(m)/m$.*

Proof. The moduli with $\phi(m) \le 2$ are $m \in \{1, 2, 3, 4, 6\}$, with $\rho(1) = 1$, $\rho(2) = \rho(4) = 1/2$, $\rho(3) = 2/3$, $\rho(6) = 1/3$. □

Theorem 6.2 (Analytic optimality)**.** *Let $\chi_0^{(6)}$ denote the principal Dirichlet character modulo 6, and let χ_{12} be the imprimitive character modulo 12 given by $\chi_{12} = \chi_{-4} \cdot \chi_0^{(3)}$, where χ_{-4} is the primitive quadratic character modulo 4. Then*

$$L(2, \chi_0^{(6)}) = [L(1, \chi_{12})]^2 = \frac{\pi^2}{9}. \tag{6.1}$$

Proof. For the principal character modulo q, the Euler product is $L(s, \chi_0^{(q)}) = \zeta(s) \prod_{p|q}(1 - p^{-s})$, so at $s = 2$ and $q = 6$,

$$L(2, \chi_0^{(6)}) = \zeta(2)(1 - 2^{-2})(1 - 3^{-2}) = \frac{\pi^2}{6} \cdot \frac{3}{4} \cdot \frac{8}{9} = \frac{\pi^2}{9},$$

using Euler's evaluation $\zeta(2) = \pi^2/6$ [22, Ch. 8]. For the second identity, the character χ_{12} vanishes at $p \in \{2, 3\}$ and equals $\chi_{-4}(p)$ for $p \geq 5$, so

$$L(1, \chi_{12}) = L(1, \chi_{-4}) \cdot (1 - \chi_{-4}(3)/3).$$

Since $3 \equiv 3 \pmod 4$, we have $\chi_{-4}(3) = -1$, hence $1 - \chi_{-4}(3)/3 = 4/3$. By the classical Leibniz evaluation $L(1, \chi_{-4}) = \pi/4$ [23, Ch. 4],

$$L(1, \chi_{12}) = \frac{4}{3} \cdot \frac{\pi}{4} = \frac{\pi}{3}.$$

Squaring, $[L(1, \chi_{12})]^2 = \pi^2/9$, which equals $L(2, \chi_0^{(6)})$. □

Both results are elementary and complete within their scope, but they do not distinguish $m = 6$ from all moduli: the algebraic optimality is a finite verification over $\{1, 2, 3, 4, 6\}$, and the analytic identity is a rational computation with the classical constants $\zeta(2)$ and $L(1, \chi_{-4})$.

Three additional optimality principles of $m = 6$ (geometric, thermodynamic, spectral) are recorded in companion work deposited on Zenodo [19], and the present paper does not rely on them. The reader interested in those developments should consult the companion preprint with the usual caution reserved for unpublished work.

6.3. **Limitations.** Four limitations of the present work are worth recording explicitly.

Complexity. The $\Theta(N^{3/2}/\log N)$ time restricts the algorithm to memory-constrained settings, not massive-scale enumeration. The classical segmented sieve of Eratosthenes runs in $O(N \log \log N)$ time and $O(\sqrt{N})$ memory, so the present algorithm is not competitive for large N; its value lies in the minimal state and in the structural analysis, not in raw speed.

Wall-clock overhead. The overhead of the asymmetric threshold implementation is a genuine limitation for time-sensitive applications up to $N = 10^{10}$. We have argued that it is an implementation artifact, but the artifact is real for the scales at which the algorithm is likely to be used.

Heuristic hybrid. The heuristic hybrid configuration (C) of Appendix C reduces the number of tests by about 13% but lacks formal correctness: the threshold $2\kappa^+_{\min}(p)$ is not a logically valid bound. The configuration is not a competitor of the asymmetric threshold, but it indicates that a carefully tuned heuristic can outperform both in practice, at the cost of losing the formal guarantee.

Growth of $j_6(p)$. The records of the entanglement jump admit two fits (power law and exponential) that are statistically indistinguishable over the accessible range, and the Hill estimator of the tail index has not stabilised by $N \sim 6 \times 10^8$. The accessible data do not determine the asymptotic growth of $j_6(p)$, and we have not claimed otherwise.

A note on the scope of the paper. The paper sits at the interface of three communities. For the analytic number theorist, its interest is the strict separation $\kappa^-_{\min} > \kappa^+_{\min}$ and the twin-prime characterisation; for the algorithmic number theorist, its interest is the minimal-state sieve and the asymptotic analysis; for the formal verification community, its interest is the Lean 4 certification of the algebraic core. We have not tried to be optimal for any single community. The primary audience is the analytic and algorithmic number theorist, and the formal verification is a supporting contribution, not a parallel narrative.

7. Conclusions

We have developed an algebraic theory of modular projection sieving over $(\mathbb{Z}/6\mathbb{Z})^\times$, with the theory of prime-coprime entanglement as its central contribution. The asymmetric thresholds $\kappa^\pm_{\min}$ strictly dominate the symmetric wheel threshold (Theorem 2.12), and the local vanishing of the entanglement jump $\Delta k(p) = 0$ is the ground state of the prime-coprime pairing (Proposition 3.3). The empirical distribution of entanglement jumps, measured at two scales with Wilson 95% confidence intervals, decreases from 15.75% to 13.83% in ground state density between $N \sim 6 \times 10^7$ and $N \sim 6 \times 10^8$, and the Lean 4 certification ensures the correctness of the algebraic core.

An ablation study shows that $\kappa^-_{\min}$ is logically optimal but operationally marginal; a heuristic hybrid configuration achieves a 13% reduction in modular tests at the cost of losing formal correctness. The structural optimality of $m = 6$ is established under algebraic and analytic principles; other optimality principles are recorded as a companion contribution, with the caveat that they are not used in the present manuscript. The algorithm runs in $\Theta(\sqrt{N}/\log N)$ working memory and matches the lower bound for state-vectorless sieves. The measurement protocol yields empirical transition ratios that match the theoretical prediction $\Theta(N^{3/2}/\log N)$ to within 5% for $N \geq 10^6$.

For the growth of the record function $j_6(p)$, the accessible data do not determine which of the two candidate asymptotic laws holds, and we have left the question open. This is the honest conclusion of the empirical analysis, and it illustrates the value of the strict measurement protocol adopted in §5.1: the protocol is designed to distinguish artifact from signal, and in the case of $j_6(p)$ the signal is below the resolution of the data.

We hope that the combination of exact computation, structural analysis, and formal verification presented here will inspire further work at the interface of arithmetic combinatorics, sieve theory, and Lean 4.

8. Open problems

(P1) *Higher moduli.* Generalise the entanglement theory to $m = 30, 210, \ldots$ and determine whether $\kappa^{-}_{\min}$ admits a closed form. The construction of the positional coprime generalises verbatim to any primorial, but the analogue of Theorem 2.12 requires a case analysis that grows with $\phi(m)$.

(P2) *Heuristic hybrid.* Determine whether the heuristic hybrid configuration (C) of Appendix C is provably correct for all N, or fails for some scale. The configuration is not logically valid in general, but no counterexample is known in the range $N \le 10^{10}$.

(P3) *Asymptotic growth of $j_6(p)$.* Determine the asymptotic growth of the record entanglement jump. Distinguishing a power law from an exponential requires primes far beyond the accessible range $N = 6 \times 10^8$; a GPU implementation or a distributed sweep may reach $N = 10^{11}$.

(P4) *Spectral properties.* Investigate the spectral properties of the entanglement operator $\hat{E}_p$ and its relation to the self-adjoint sieve operator $\mathbf{H}_N = MM^T$. The connection is the subject of a companion preprint [20] and is not developed here.

(P5) *Acyclicity of entanglement chains.* Determine whether the entanglement chains of Construction 3.1 are infinite for every root $p_0 \in \mathcal{P}_{>3}$, i.e., whether Cop is acyclic. No cycle has been observed up to $N = 6 \times 10^8$; a proof of acyclicity would settle the infiniteness of the chains unconditionally (currently conditional on Dirichlet and on the absence of cycles).

Appendix A. Characterisation of the composite index set

The following proposition is auxiliary to the structural analysis of §3; the proof of Theorem 3.2 reduces to it.

Proposition A.1 (Structure of the composite index set)**.** *Let $\mathcal{K}_{\text{comp}} \subset \mathbb{N}^+$ be the set of indices k such that at least one of $6k-1$, $6k+1$ is composite. Then*

(A.1)

$$\mathcal{K}_{\text{comp}} = \bigcup_{p \in \mathcal{P}_{>3}} \Big(\{k \in \mathbb{N}^+ : k \equiv k_p \pmod p,\ k > k_p\} \cup \{k \in \mathbb{N}^+ : k \equiv -k_p \pmod p\} \Big).$$

Proof. Write S_p for the set in braces on the right-hand side of (A.1), and let $\mathcal{S} = \bigcup_{p \in \mathcal{P}_{>3}} S_p$.

Forward inclusion $\mathcal{K}_{\text{comp}} \subseteq \mathcal{S}$. Let $k \in \mathcal{K}_{\text{comp}}$, so that $6k + \varepsilon$ is composite for some $\varepsilon \in \{1, -1\}$. Let p be the smallest prime factor of $6k + \varepsilon$, and write $6k + \varepsilon = pq$ with $q \ge p$. Since $6k + \varepsilon \equiv \varepsilon \pmod 6$ and $\varepsilon^2 = 1$, the integer $6k + \varepsilon$ is odd and not divisible by 3; hence $p \in \mathcal{P}_{>3}$ and $q \ge p \ge 5 > 1$. So p is a proper factor, and in particular $6k + \varepsilon \ne p$.

By Lemma 2.4 applied to $N = 6k + \varepsilon$, the divisibility $p \mid N$ is equivalent to $k \equiv \varepsilon_p \varepsilon k_p \pmod{p}$.

If $\varepsilon_p = \varepsilon$ (the base prime p lies in the same channel as the candidate), this reduces to $k \equiv k_p \pmod{p}$. We claim $k > k_p$: since $6k + \varepsilon \neq p$ and both sides are positive multiples of the same congruence structure, the only alternative is $6k + \varepsilon < p$; but $6k + \varepsilon < p$ would contradict $p \mid (6k + \varepsilon)$ unless $6k + \varepsilon = 0$, which is impossible for $k \geq 1$. Hence $k > k_p$, and k lies in the first branch of S_p.

If $\varepsilon_p = -\varepsilon$ (opposite channel), the congruence is $k \equiv -k_p \pmod{p}$, and k lies in the second branch of S_p. In either case $k \in \mathcal{S}$.

Backward inclusion $\mathcal{S} \subseteq \mathcal{K}_{\text{comp}}$. Let $k \in S_p$ for some $p \in \mathcal{P}_{>3}$. Two cases.

Case A: $k \equiv k_p \pmod{p}$ and $k > k_p$. If $\varepsilon_p = +1$ ($p \in \mathcal{C}_1$), Lemma 2.4 with $\varepsilon = +1$ gives $p \mid (6k+1)$; since $6k+1 \geq 6(k_p+1)+1 = p+6 > p$, the integer $6k + 1$ has the proper divisor p, hence is composite. If $\varepsilon_p = -1$ ($p \in \mathcal{C}_5$), Lemma 2.4 with $\varepsilon = -1$ gives $p \mid (6k - 1)$; since $6k - 1 \geq 6(k_p + 1) - 1 = p + 6 > p$, the integer $6k - 1$ is composite.

Case B: $k \equiv -k_p \pmod{p}$. If $\varepsilon_p = +1$ ($p \in \mathcal{C}_1$), Lemma 2.4 with $\varepsilon = -1$ gives $p \mid (6k - 1)$. If $6k - 1 = p$, then $6k - 1 = 6k_p + 1$, i.e., $6(k - k_p) = 2$, which has no solution in integers; hence $6k - 1 > p$, and $6k - 1$ is composite. If $\varepsilon_p = -1$ ($p \in \mathcal{C}_5$), Lemma 2.4 with $\varepsilon = +1$ gives $p \mid (6k+1)$. If $6k+1 = p$, then $6k+1 = 6k_p-1$, i.e., $6(k-k_p) = -2$, again impossible; hence $6k+1 > p$, and $6k + 1$ is composite.

In every case, at least one of $6k - 1$, $6k + 1$ is composite, so $k \in \mathcal{K}_{\text{comp}}$. Combining the two inclusions, $\mathcal{K}_{\text{comp}} = \mathcal{S}$. □

Remark A.2 (Role of the $k > k_p$ exclusion)**.** The exclusion $k > k_p$ in the first branch of (A.1) is essential: at $k = k_p$ the candidate $6k+\varepsilon = 6k_p+\varepsilon_p = p$ equals the base prime itself and is prime, not composite. The second branch $\{k \equiv -k_p \pmod{p}\}$ requires no such exclusion, because $k \equiv -k_p$ forces $k \geq p - k_p = 5k_p \pm 1 > k_p$ for $k \geq 1$: the smallest positive solution is already strictly greater than k_p.

Appendix B. Pseudocode

Appendix C. Ablation study

We compare three configurations of the threshold assignment in the algorithm of Appendix B:

(A) **Symmetric**: use $\kappa^{+}_{\min}(p)$ in both channels. Minimal state, no auxiliary structure.
(B) **Asymmetric**: use $\kappa^{\pm}_{\min}(p)$ via Definition 2.7. Requires the NextKop pointer and the $\mathcal{K}^{-}_{\min}$ array.
(C) **Heuristic hybrid**: use $\kappa^{+}_{\min}(p)$ in the own channel and $2\kappa^{+}_{\min}(p)$ in the opposite channel. No NextKop pointer.

Three observations follow from Table 4.

Algorithm 1 Modular projection sieve with $\kappa^{\pm}_{\min}$ entanglement

Require: $k_{\max}$
Ensure: C = number of primes $\leq 6k_{\max}+1$
$\mathcal{P}_{\text{base}} \leftarrow [\,]$; $\mathcal{K}_p \leftarrow [\,]$; $\mathcal{E}_p \leftarrow [\,]$
$\mathcal{K}^{+}_{\min} \leftarrow [\,]$; $\mathcal{K}^{-}_{\min} \leftarrow [\,]$; $\mathcal{N}_{\text{kop}} \leftarrow [\,]$
$C \leftarrow 2$; size $\leftarrow 0$; $N_{\max} \leftarrow 6k_{\max}+1$
for $k \leftarrow 1$ **to** $k_{\max}$ **do**
  for $\varepsilon \leftarrow \{-1,+1\}$ **do**
    if IsCandidate$(k,\varepsilon,\ldots)$ **then**
      $C \leftarrow C+1$
      if $(6k+\varepsilon)^2 \leq N_{\max}$ **then**
        $k_{\text{kop}} \leftarrow$ NextKop$(k,\varepsilon,\ldots)$
        $(\kappa^{+},\kappa^{-}) \leftarrow$ KminPM$(6k+\varepsilon,k,\varepsilon,k_{\text{kop}})$
        Append$(6k+\varepsilon,k,\varepsilon,\kappa^{+},\kappa^{-},k_{\text{kop}})$
      end if
    end if
  end for
end for
return C

Algorithm 2 IsCandidate$(k,\varepsilon,\ldots)$

Apply the digital filter modulo 10 (Theorem 4.1)
for $i \leftarrow 0$ **to** size -1 **do**
  if $\varepsilon=-1$ **and** $k<\mathcal{K}^{-}_{\min}[i]$ **then**
    continue
  end if
  if $\varepsilon=+1$ **and** $k<\mathcal{K}^{+}_{\min}[i]$ **then**
    continue
  end if
  if $\varepsilon=\mathcal{E}_p[i]$ **and** $(k-\mathcal{K}_p[i]) \bmod \mathcal{P}_{\text{base}}[i]=0$ **then**
    return false
  end if
  if $\varepsilon\neq\mathcal{E}_p[i]$ **and** $(k+\mathcal{K}_p[i]) \bmod \mathcal{P}_{\text{base}}[i]=0$ **then**
    return false
  end if
end for
return true

First, configuration (B) is logically optimal but operationally marginal: the reduction in the number of tests is between 0.006% and 0.047%, and decreases with N as predicted by the theoretical analysis of §4.7. The wall-clock overhead at $N=10^{10}$ is +0.50%, larger than the savings in tests; this is the artifact discussed in §6.1.

TABLE 4. Ablation of the threshold configuration. Tests are counted analytically (exact, no approximation). The relative reductions are computed with respect to configuration (A).

N	Tests (A)	Tests (B)	Tests (C)	$\Delta_{B/A}$	$\Delta_{C/A}$
10^8	2.838×10^{10}	2.837×10^{10}	2.467×10^{10}	0.047%	13.07%
10^9	7.850×10^{11}	7.849×10^{11}	6.810×10^{11}	0.017%	13.25%
10^{10}	2.203×10^{13}	2.202×10^{13}	1.907×10^{13}	0.006%	13.40%

Second, configuration (C) reduces the number of tests substantially (13%) but lacks formal correctness: the threshold $2\kappa^+_{\min}(p)$ is not a logically valid bound. No composite is lost in practice up to $N = 10^{10}$, because the composites with $q < 2p$ have already been sieved by q. Whether this empirical correctness persists for larger N is open problem (P2) in §8.

Third, the trade-off between (B) and (C) is structural, not pathological: $\kappa^-_{\min}$ is a structural concept, not an algorithmic one. Configuration (B) is the correct implementation of the theory; configuration (C) is a heuristic that may be of practical interest but does not belong to the theory.

APPENDIX D. LEAN 4 CERTIFICATION SUMMARY

The algebraic results of §2, §3, and §6.2 are certified in Lean 4 [13] (v4.35.0, Mathlib4 [14]) with zero `sorry` gaps and zero global `axiom` declarations. The suite comprises nineteen self-contained modules, which we classify along two levels of certainty.

Certification levels. Level *N1* denotes a theorem whose proof depends only on Lean's kernel, Mathlib's foundational axioms, and no external analytical input; it is certified unconditionally. Level *N1-P* (*parametric*) denotes a theorem whose proof is unconditional in Lean but whose statement contains an explicit hypothesis corresponding to an external analytical result. The distinction prevents axiomatic pollution: a global `axiom` declaration would silently import the analytical input into every downstream proof, whereas a parameterised hypothesis keeps the dependency visible, local, and falsifiable. No module of the suite reaches the weaker level *N2* (containing `sorry` or a global `axiom`).

In the present suite, seventeen modules are certified at level N1 and two at level N1-P. The two N1-P modules are:

- `CopExistence`, whose statement is conditional on the existence of a prime in the opposite channel above a given index—a consequence of Dirichlet's theorem [21].
- `AnalyticOptimality`, whose statement is conditional on the two classical evaluations $\zeta(2) = \pi^2/6$ and $L(1, \chi_{-4}) = \pi/4$ of [22, 23].

D.1. **Module table.**

TABLE 5. The Lean 4 suite. Modules are grouped by the manuscript section they certify. N1: pure algebraic; N1-P: parametric, with external analytical hypotheses localised in the theorem signature.

Module	Ref.	Certified statement	Level
§2. Algebraic framework			
`ModularClassification`	Thm. 2.1	Modular classification (4 classes) + prime coverage $p > 3$	N1
`UnitGroupStructure`	§2.1	$(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$; chiral involution	N1
`PositionalCongruence`	Lem. 2.4	Positional congruence (Euclid via Bézout)	N1
`KminPlus`	Thm. 2.6	$\kappa^+_{\min}$ formula	N1
`EntanglementThresholds`	Thm. 2.9	$\kappa^-_{\min}$ formula (4 chiral cases)	N1
`CopExistence`	Def. 2.7, Thm. 2.8	$\mathrm{Cop}(p)$ existence (Dirichlet param.)	N1-P
`ThresholdSeparation`	Thm. 2.12	$\kappa^-_{\min}(p) > \kappa^+_{\min}(p)$ strictly	N1
§3 & App. A. Structural consequences and twin primes			
`CompositeStructure`	Prop. A.1	Local branching structure of composite indices	N1
`TwinPrimeGroundState`	Thm. 3.2	Twin-prime index (full equivalence)	N1
`TwinPrimesBoth`	Rem. 2.10	Mono-prime incompatibility; bilateral chirality	N1
`TwinPrimesExt`	§3.3	HL local identity + density factorization	N1
`HardyLittlewood`	Prop. 3.5	HL local factor $= 1 - (p-1)^{-2}$	N1
`EulerFactorization`	§2.2	Finite Euler factorization over $(\mathbb{Z}/6\mathbb{Z})^\times$	N1
§4–5. Sieve algorithm and operators			
`DigitalFilter5`	Thm. 4.1	Modulo 5 filter (and mod 10 corollary)	N1
`Complexity`	Thms. 4.2–4.3	Memory/time bounds (parametric)	N1
`SieveOperatorSelfAdjoint`	§4.3	$(MM^T)^T = MM^T$	N1
`ChiralConvergence`	§5	Ratio identity c_1/c_5	N1
§6.2. Structural optimality of $m = 6$			
`OptimalityM6`	§6.2	$\rho(m)$ minimised at $m = 6$	N1
`AnalyticOptimality`	Thm. 6.2	$L(2, \chi_0^{(6)}) = [L(1, \chi_{12})]^2 = \pi^2/9$ (param.)	N1-P

D.2. **Consolidated certification remarks.** The following remarks, collected from the body of the paper, summarise the correspondence between manuscript statements and Lean 4 modules.

Unit group and chiral involution. The identity $(\mathbb{Z}/6\mathbb{Z})^\times = \{1, 5\}$ and the involution $5 \equiv -1 \pmod 6$ (both multiplicatively and additively) are certified `sorry`-free in module `UnitGroupStructure` at level N1. They underlie the two-channel decomposition of Definition 2.7 and the asymmetry of Theorem 2.12.

Positional congruence. Lemma 2.4 is certified `sorry`-free in module `PositionalCongruence` at level N1. The proof is factored through an auxiliary Euclidean lemma (`dvd_of_dvd_mul_6`) that reduces the divisibility $p \mid X$ from $p \mid 6X$ to Bézout's identity $\gcd(6, p) = 1$. The formal proof is not a Presburger identity: it invokes `IsCoprime`, `linear_combination` and `ring`, not merely `omega`. The manuscript proof ("Since $\gcd(6, p) = 1 \ldots$") and the Lean proof are structurally isomorphic.

Local branching structure. The local form of Proposition A.1 is certified `sorry`-free in module `CompositeStructure` at level N1: for every $p \in \mathcal{P}_{>3}$ and every candidate $6k + \varepsilon$,

$$p \mid (6k{+}\varepsilon) \wedge 6k{+}\varepsilon \neq p \iff \big(\varepsilon = \varepsilon_p \wedge p \mid (k{-}k_p) \wedge k > k_p\big) \vee \big(\varepsilon = -\varepsilon_p \wedge p \mid (k{+}k_p)\big).$$

The *global* set-theoretic identity of Proposition A.1—that $\mathcal{K}_{\text{comp}}$ equals the union of the local branches over all $p \in \mathcal{P}_{>3}$—follows by a standard union argument and is not formalised separately.

Threshold formulas. The $\kappa^+_{\min}$ formula (Theorem 2.6) is certified `sorry`-free in module `KminPlus` at level N1. The four chiral cases of the $\kappa^-_{\min}$ formula (Theorem 2.9) are certified `sorry`-free in module `EntanglementThresholds` at level N1; each case is a pure polynomial identity in $\mathbb{Z}$, discharged by `ring`.

Strict threshold separation. Theorem 2.12 is certified `sorry`-free in module `ThresholdSeparation` at level N1, via `omega`, `nlinarith` and `ring`. Both chiral cases are proved as separate lemmas (`kmin_minus_gt_kmin_plus_C5` and `kmin_minus_gt_kmin_plus_C1`); the auxiliary inequality $p^2 < pq$ for $0 < p < q$ is certified as `kmin_plus_lt_pq`.

Twin-prime index characterization. Theorem 3.2 is certified `sorry`-free in module `TwinPrimeGroundState` at level N1. The formalisation establishes the equivalence in both directions: necessity follows from the definition of primality and the factorisation of $6k{\pm}1$; sufficiency uses `Nat.minFac` and the classical bound $\text{minFac}(n)^2 \leq n$ for composite n. The formalisation confirms that Theorem 3.2 is not conditional on any analytical hypothesis: it is a pure statement about the arithmetic of $\mathbb{Z}/6\mathbb{Z}$ and the classical characterisation of prime numbers via minimal factors.

Hardy–Littlewood local factor. Proposition 3.5 is certified `sorry`-free in module `HardyLittlewood` at level N1. The finite Euler factorisation underlying the proposition (restricted to primes in $\mathcal{C}_1 \cup \mathcal{C}_5$) is certified in module `EulerFactorization` at level N1. The *passage to the infinite product* that appears in the Hardy–Littlewood prediction is not formalised, and this is the reason Proposition 3.5 is stated conditionally on the CRT independence hypothesis.

Digital filter modulo 5. Theorem 4.1 is certified `sorry`-free in module `DigitalFilter5` at level N1. The key identities $5 \mid (6k+1)-(k+1)$ and $5 \mid (6k-1)-(k-1)$ are proved by `omega`, and the iff-forms are derived from them by `dvd_add` and `dvd_sub`.
Sieve operator self-adjointness. The symmetry $(MM^T)^T = MM^T$ of the sieve operator is certified `sorry`-free in module `SieveOperatorSelfAdjoint` at level N1, for any matrix M over a commutative ring.
Analytic optimality. The algebraic core of Theorem §6.2 is certified `sorry`-free in module `AnalyticOptimality` at level N1-P. Three N1 lemmas on the local factors ($(1-2^{-2}) = 3/4$, $(1-3^{-2}) = 8/9$, and their product $2/3$) are certified in the same module; the N1-P statement (`opt_analytic_identity`) contains the two classical evaluations as explicit hypotheses.
Incompatibility and bilateral structure. The mono-prime incompatibility (a prime $p \geq 5$ cannot divide $6k+1$ and $6k-1$ simultaneously), the bilateral chiral involution, and the hierarchy of bilateral runs are certified `sorry`-free in module `TwinPrimesBoth` at level N1. This module also certifies the elementary asymptotic fact that the primality window $X(z) = z^2/6$ eventually dominates any linear bound Cz.
Density factorization. The local Hardy–Littlewood identity $(p-2)/p = (1-1/p)^2\,(1-1/(p-1)^2)$ for $p \geq 5$, the factorization of the survival density into the coprime density times the Hardy–Littlewood factor, the positivity of each factor for every finite scale $z \geq 5$, and the multiplicative relation $M(z)\cdot\text{survival} = \prod(p-2)$ are certified `sorry`-free in module `TwinPrimesExt` at level N1. The module certifies the algebraic core of Proposition 3.5 unconditionally; the passage to the limit $z \to \infty$ is not formalised, which is why Proposition 3.5 remains stated conditionally on the CRT independence hypothesis.
Reproducibility. The nineteen-module suite is provisioned via `elan`, with strict toolchain synchronisation against Mathlib4, cache retrieval (`lake exe cache get!`), and per-file kernel audit (`lake env lean <file>`). Each module is accompanied by a SHA-256 digest, and the complete suite is packaged as a reproducible ZIP archive. The reproducibility manifest, the per-module mapping to manuscript statements, and the exact compiler versions are provided in the companion repository listed in §Data and Code Availability.

Declarations

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Author Contributions. The sole author (José Ignacio Peinador Sala) conceived the theoretical framework of modular projection sieving over $(\mathbb{Z}/6\mathbb{Z})^\times$, derived and proved the asymmetric activation thresholds, developed and formally verified the 19-module suite in Lean 4, implemented the Python/Numba sieve engine, designed and executed the empirical benchmarking protocol, and wrote the manuscript in its entirety.

AI Usage Statement. The author declares having used generative artificial intelligence assistance **Gemini (Google)** and **Deepseek (High-Flyer)** exclusively during manuscript preparation for typographical formatting assistance (LaTeX syntax, table alignment, and cross-referencing), language translation, and stylistic-linguistic editing. The mathematical theory, formal statements, mechanized proofs in Lean 4, sieve algorithms, computational implementations, and data analyses were entirely conceived, verified, and audited by the author, who assumes full epistemic responsibility for the rigor and integrity of the published work.

Data and Code Availability. In accordance with the computational reproducibility standards of *Mathematics of Computation*, the complete computational suite supporting this manuscript has been deposited open-access in Zenodo under the permanent DOI:

`https://doi.org/10.5281/zenodo.22043293`

The repository comprises two primary artefacts:

(i) **Lean 4 verification suite:** Nineteen self-contained modules (seventeen Level N1 and two Level N1-P) certifying all algebraic results of §2–3 and §6.2 with zero `sorry` gaps and zero global `axiom` declarations, together with the environment configuration files, per-file SHA-256 digests, and build manifests.

(ii) **Experimental validation suite:** The Python/Numba source code used to produce Tables 1–3 and the record statistics of §5.5, including the raw timing logs, bootstrap and Wilson confidence interval scripts, and SHA-256 checksums of all generated numerical outputs.

The Lean 4 suite is fully self-contained and relies only on the foundational kernel of Lean 4 and Mathlib4.

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